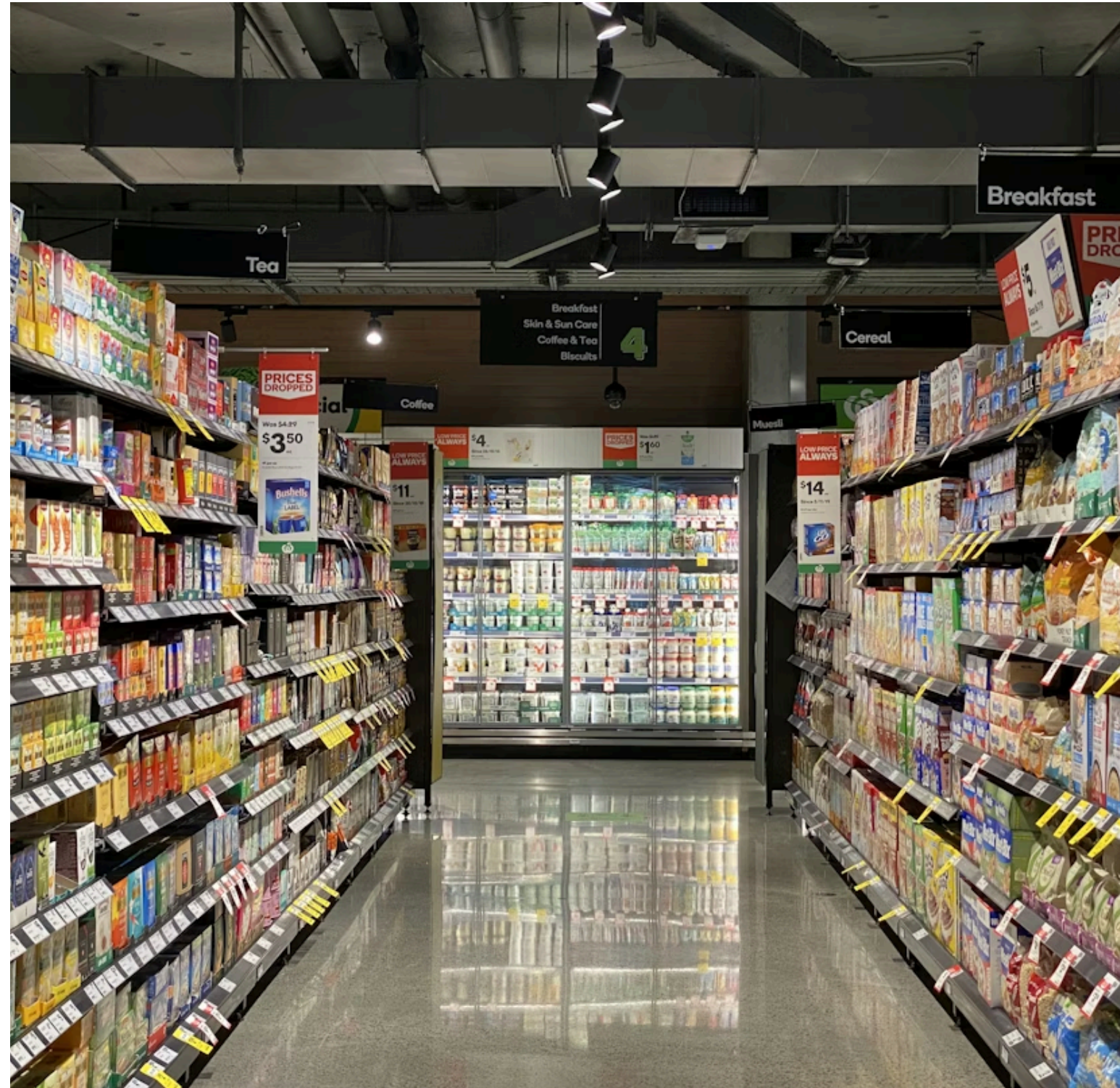




PROJECT 1

SALES TRANSACTION

Statistic for Business

PRESENTED BY MUNAWAR

BACKGROUND

This dataset summarizes sales transactions in New York City throughout 2024, covering business activities across the boroughs of Queens, Brooklyn, and Manhattan. The data provides detailed sales breakdowns by product line and customer segmentation to support the analysis of consumer behavior across various urban areas.

VARIABLE DESCRIPTION

Date	: Transaction date	Product Line	: Product Category
Branch	: Store location	Unit Price	: Price per item
Customer Type	: Customer status	Quantity	: Number of items purchased
Gender	: Customer gender	Payment	: Payment method
Rating	: Customer satisfaction		

KEY OBJECTIVE

Our objective is to analyze this data to generate a detailed performance report.:

1. The average revenue generated per transaction and total quantity of products sold.
2. Identification of the transaction with the highest purchase volume.
3. The distribution of the quantity, price, and total revenue variables.
4. The correlation between price and quantity.
5. The region with the highest number of transactions.
6. What are the most frequently purchased products?
7. Which region generates the highest and lowest total revenue?
8. What was the sales trend over the months?
9. Which products contribute the most to the total transaction earnings?
10. Based on your findings, what strategies would you recommend to the business to maximize profit?

LOAD THE DATASET

	Date	Branch	Customer type	Gender	Product line	Unit price	Quantity	Payment	Rating
0	1/1/2024	Brooklyn	Member	Female	Food & Beverages	84.63	10	Credit card	9.0
1	1/1/2024	Queens	Normal	Female	Electronics	63.22	2	Cash	8.5
2	1/1/2024	Brooklyn	Normal	Female	Electronics	74.71	6	Cash	6.7
3	1/1/2024	Queens	Member	Female	Sports & Travel	36.98	10	Credit card	7.0
4	1/1/2024	Manhattan	Member	Female	Sports & Travel	27.04	4	Ewallet	6.9

CHECKING THE MISSING VALUE

```
Date          0
Branch        0
Customer type 0
Gender        0
Product line  0
Unit price    0
Quantity      0
Payment       0
Rating        0
dtype: int64
```

There is no missing value in data table, we don't need any missing data handling process

INFORMATION THE DATASET

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5053 entries, 0 to 5052
Data columns (total 9 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Date            5053 non-null   object
1   Branch          5053 non-null   object
2   Customer type   5053 non-null   object
3   Gender          5053 non-null   object
4   Product line    5053 non-null   object
5   Unit price      5053 non-null   float64
6   Quantity        5053 non-null   int64
7   Payment         5053 non-null   object
8   Rating          5053 non-null   float64
dtypes: float64(2), int64(1), object(6)
memory usage: 355.4+ KB
```

- The dataset has 9 columns and 5,053 rows
- We will add a new column 'TotalMoney', to represent the total revenue generated from each transaction.

	Date	Branch	Customer type	Gender	Product line	Unit price	Quantity	Payment	Rating	TotalMoney
0	1/1/2024	Brooklyn	Member	Female	Food & Beverages	84.63	10	Credit card	9.0	846.30
1	1/1/2024	Queens	Normal	Female	Electronics	63.22	2	Cash	8.5	126.44
2	1/1/2024	Brooklyn	Normal	Female	Electronics	74.71	6	Cash	6.7	448.26
3	1/1/2024	Queens	Member	Female	Sports & Travel	36.98	10	Credit card	7.0	369.80
4	1/1/2024	Manhattan	Member	Female	Sports & Travel	27.04	4	Ewallet	6.9	108.16

DESCRIPTIVE STATISTICS

	Unit price	Quantity	TotalMoney
count	5053.000000	5053.000000	5053.000000
mean	39.692988	4.593509	186.531734
std	29.557474	2.787934	200.196697
min	2.010000	1.000000	2.180000
25%	11.630000	2.000000	38.520000
50%	33.900000	4.000000	109.600000
75%	64.940000	7.000000	266.560000
max	99.960000	10.000000	993.000000

- We can use descriptive statistics to see how diverse the data is for each numerical variable.

1.THE AVERAGE REVENUE GENERATED PER TRANSACTION AND TOTAL QUANTITY OF PRODUCTS SOLD.

- The average transaction in 2024 was \$186.53, with a maximum of \$993 and a minimum of \$2.18
- and the average quantity per transaction was 4 units, ranging from maximum of 10 and a minimum of 1

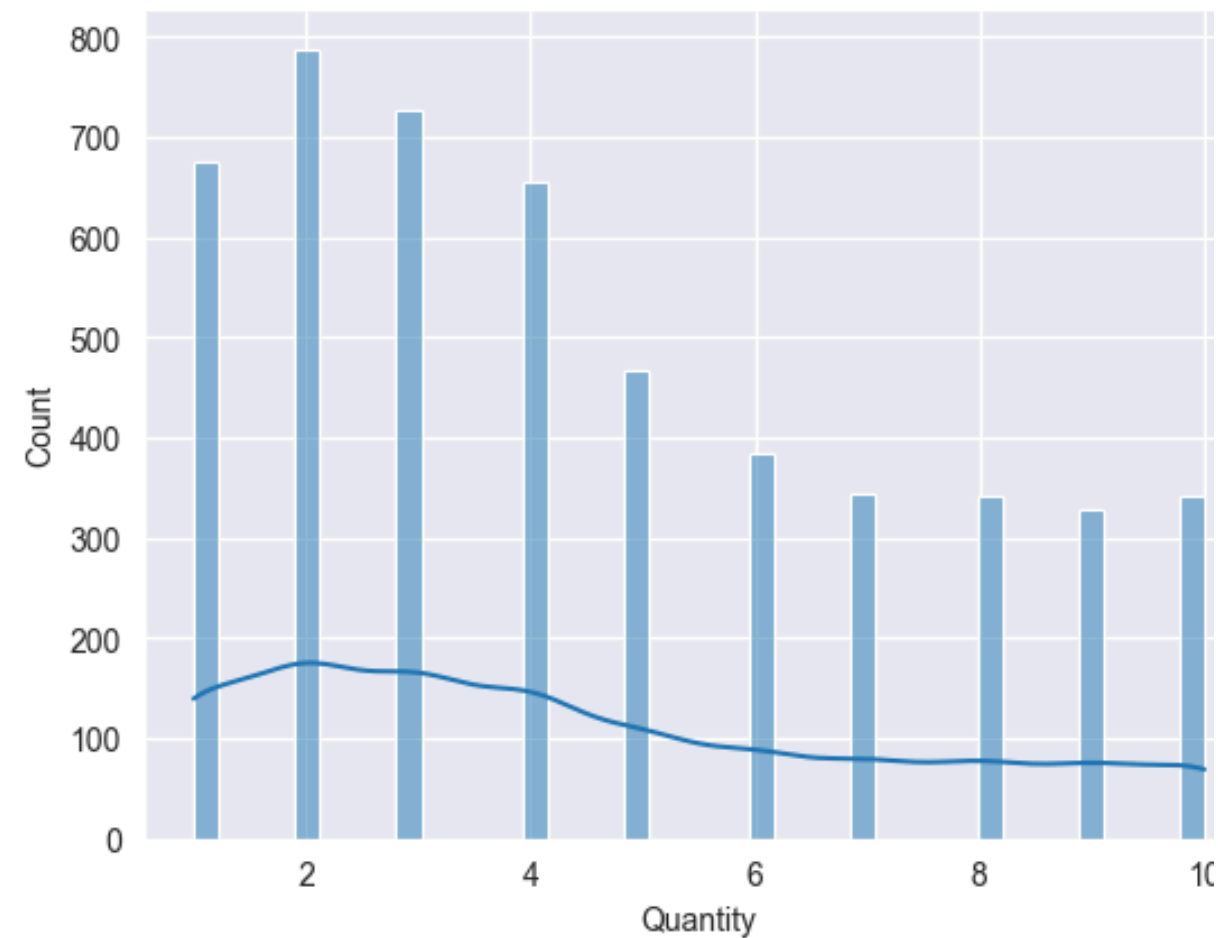
2.IDENTIFICATION OF THE TRANSACTION WITH THE HIGHEST PURCHASE VOLUME.

- The product sold in 2024 was have average \$39.69, with a maximum of \$99.96 and a minimum of \$2.01

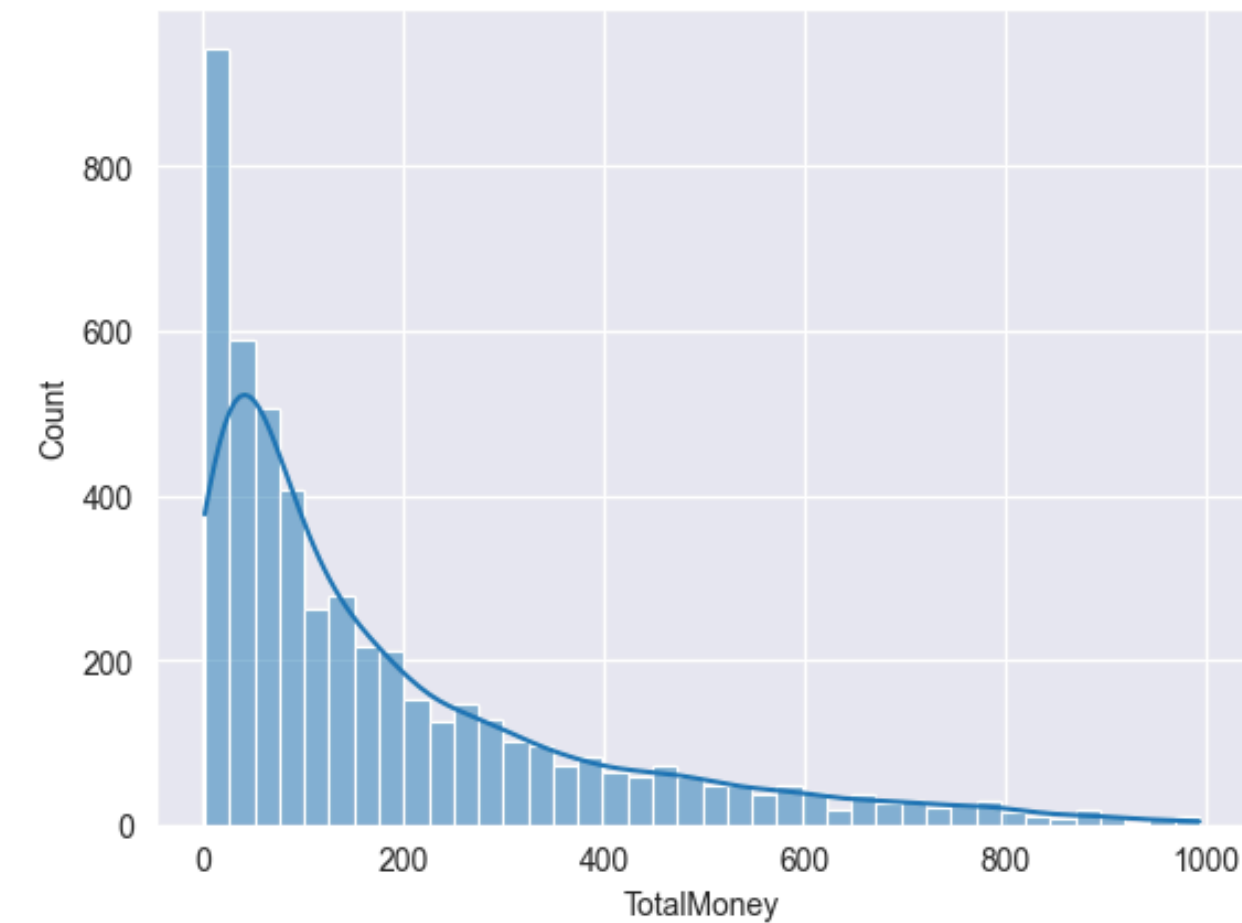
	Date	Branch	Customer type	Gender	Product line	Unit price	Quantity	Payment	Rating	TotalMoney
233	1/23/2024	Queens	Normal	Male	Health & Beauty	99.96	7	Cash	6.1	699.72
768	3/9/2024	Brooklyn	Member	Male	Sports & Travel	99.96	9	Credit card	4.2	899.64
1183	4/9/2024	Brooklyn	Member	Male	Sports & Travel	99.96	9	Credit card	4.2	899.64

- There was 3 maximum price transactions, one was located in Queens, while the other two were in Brooklyn

3. THE DISTRIBUTION OF THE QUANTITY, PRICE, AND TOTAL REVENUE VARIABLES



- The most purchases occurred in the amount of 2 units (close to 800 transactions), followed by 3 units and 1 unit.
- It seems that the trend is that the more items there are in a transaction, the lower the frequency.

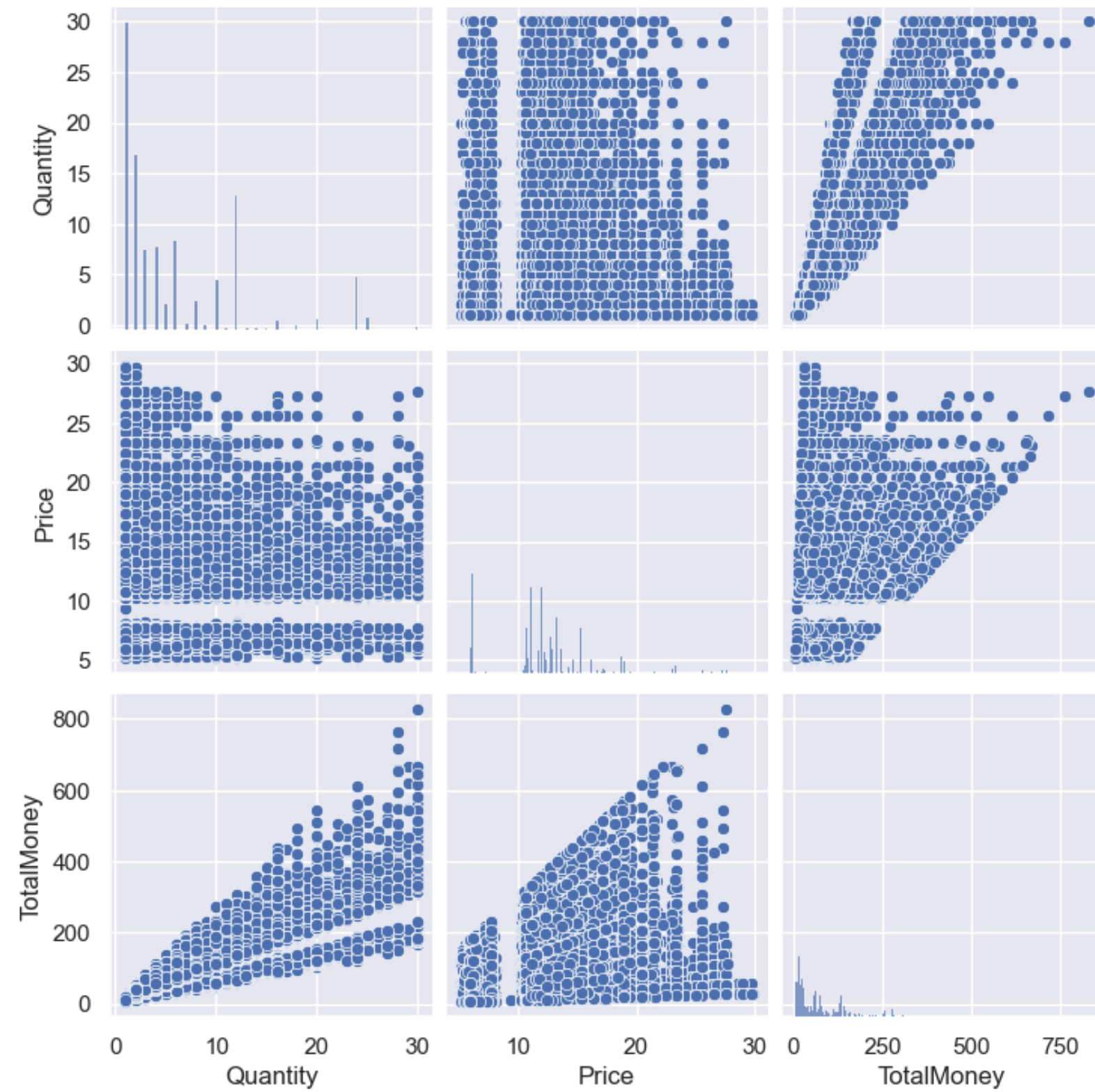


- The distribution of TotalMoney is heavily right-skewed, indicating that the vast majority of transactions are concentrated in the lower price range

4. THE CORRELATION BETWEEN PRICE AND QUANTITY

	Unit price	Quantity
Unit price	1.000000	0.050998
Quantity	0.050998	1.000000

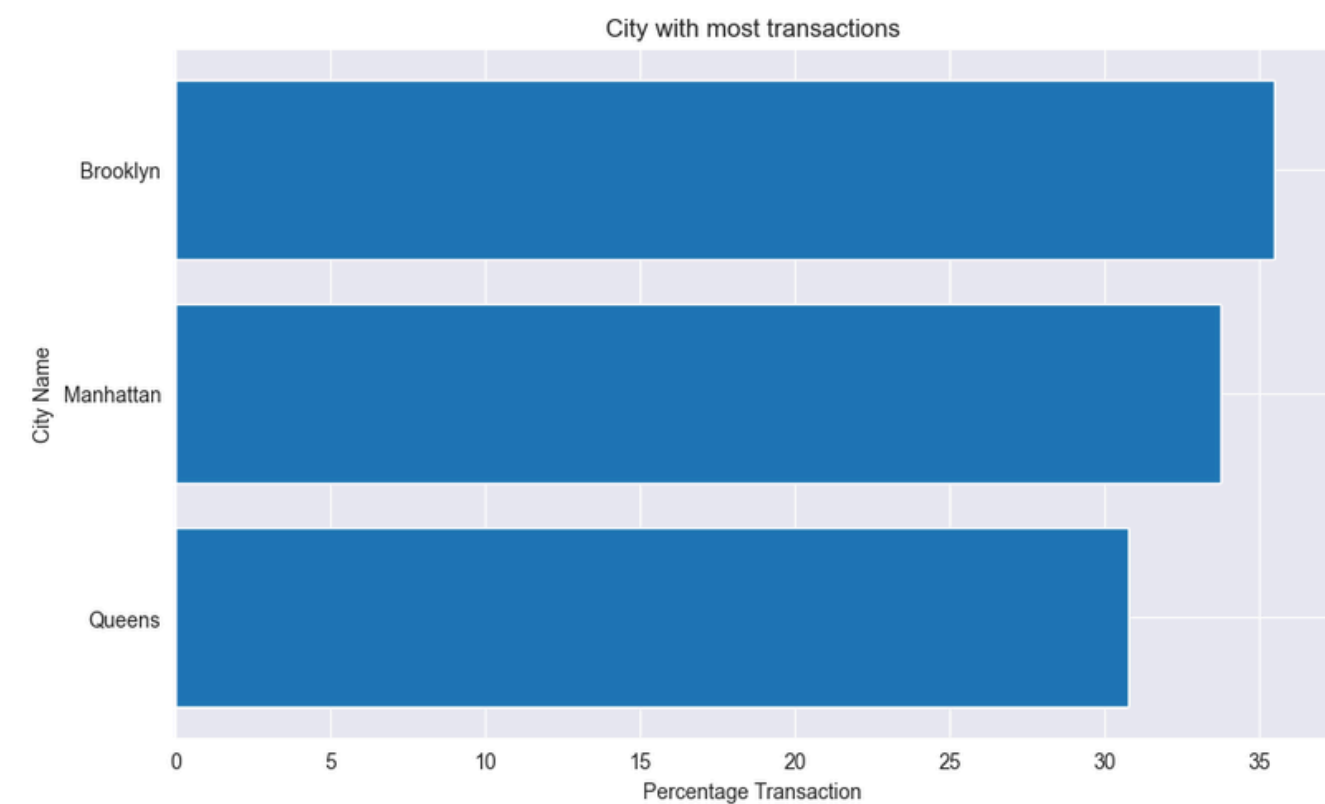
- The correlation value shows a coefficient of 0.051 between Unit price and Quantity. This indicates a **weak correlation**, suggesting that the unit price has little to no linear impact on the quantity purchased per transaction.
- we can to visualize a histogram and scatterplot of more than 2 numeric variables in a dataframe to makes it easier to see the relationship between variables



5.THE REGION WITH THE HIGHEST NUMBER OF TRANSACTIONS

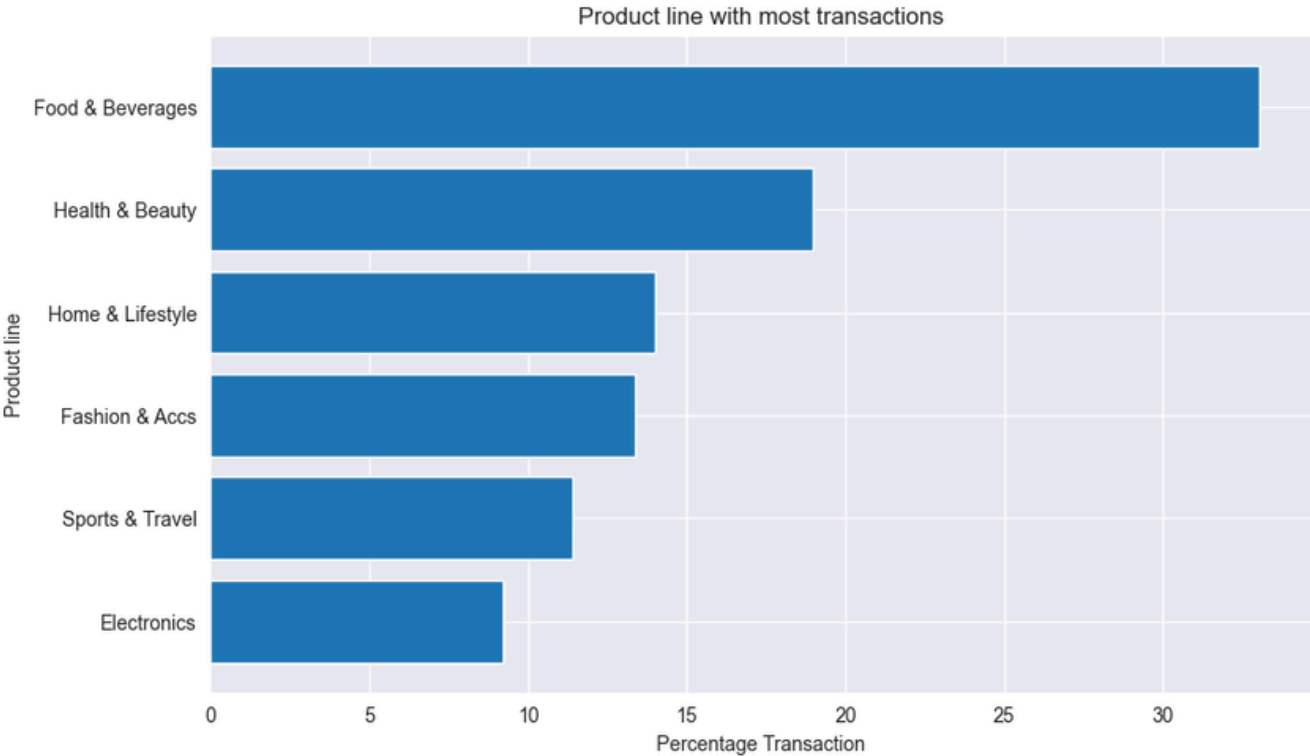
- Brooklyn stands as the leading contributor, accounting for 35.48% of the total transactions, closely followed by Manhattan at 33.74% and Queens at 30.77%. The relatively balanced distribution across these three cities indicates a stable and diversified market presence throughout the region.

	City	Percentage Transaction
0	Brooklyn	35.483871
1	Manhattan	33.742331
2	Queens	30.773798



6.THE MOST FREQUENTLY PURCHASED PRODUCTS

	Product line	Percentage Transaction
0	Food & Beverages	33.049673
1	Health & Beauty	18.959034
2	Home & Lifestyle	14.011478
3	Fashion & Accs	13.378191
4	Sports & Travel	11.399169
5	Electronics	9.202454

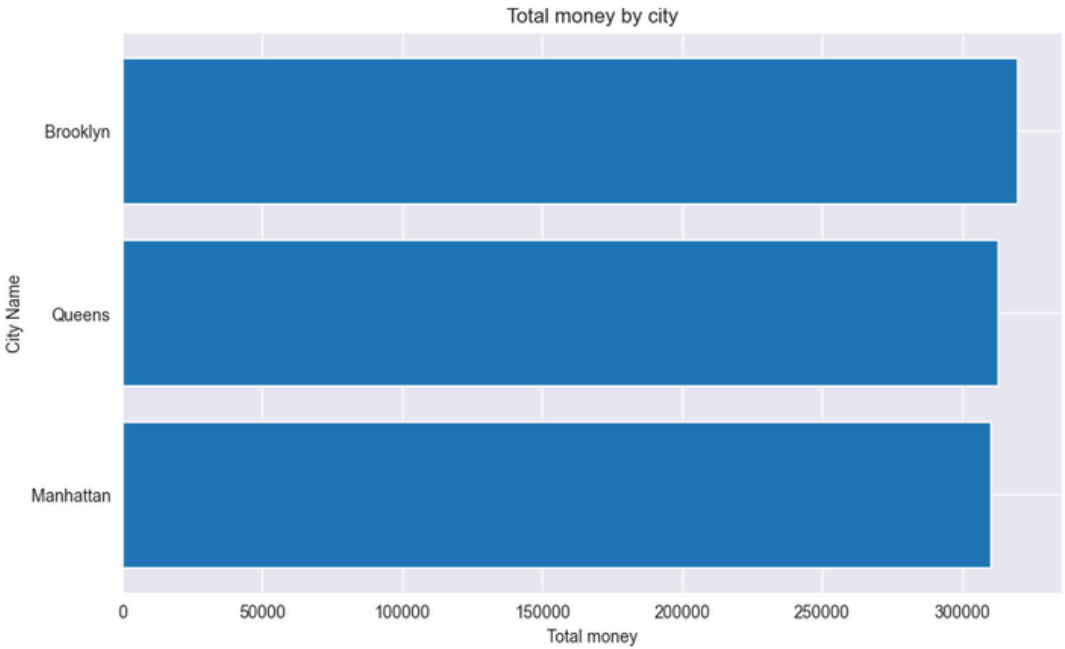


- Food & Beverages is the primary driver of transaction volume, accounting for 33.05% of all records. Health & Beauty follows as the second most active category at 18.96%. In contrast, Electronics records the lowest transaction frequency at 9.20%, which is typical for high-value, low-frequency purchase items

7. REGION GENERATES THE HIGHEST AND LOWEST TOTAL REVENUE

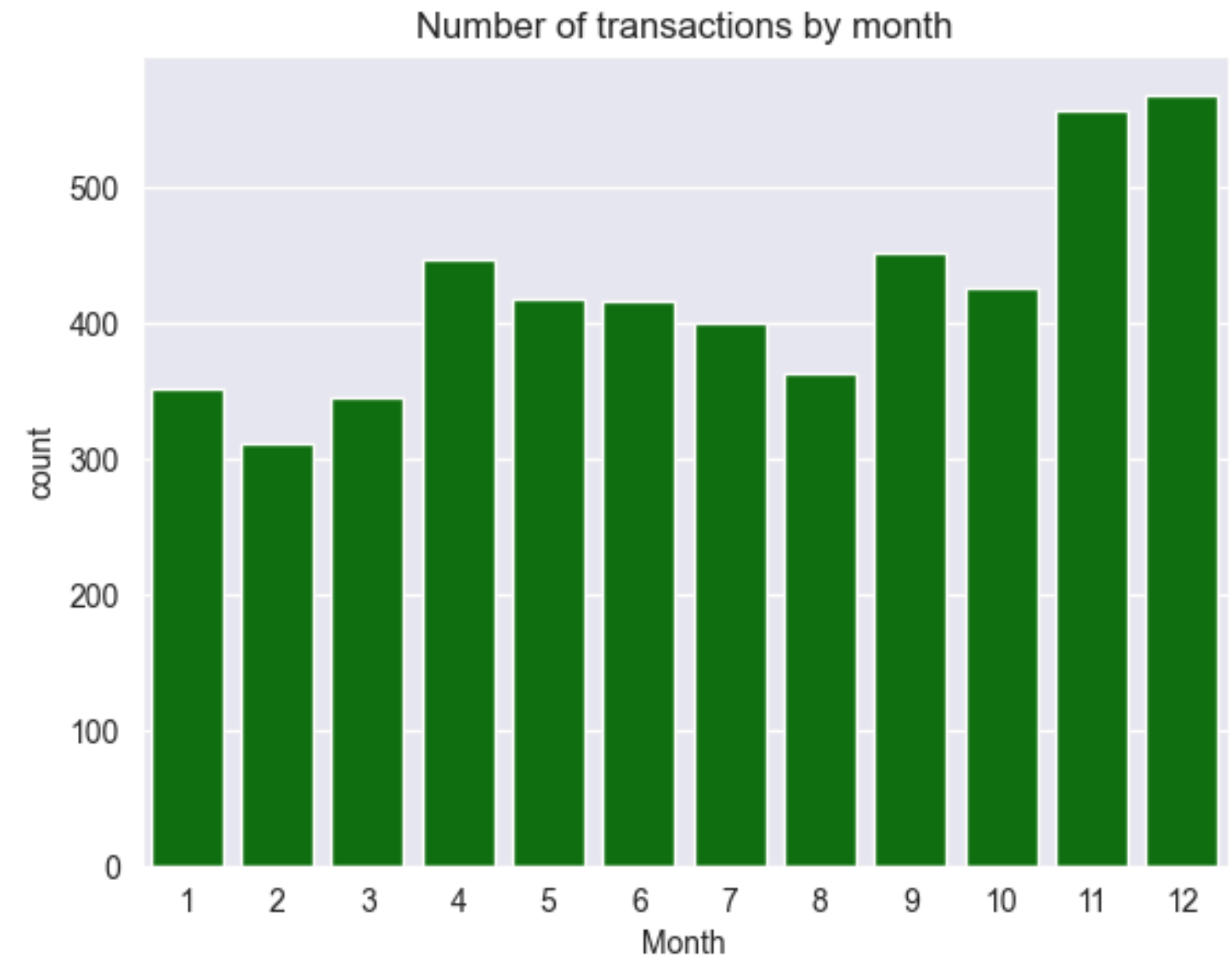
- The total revenue across the three branches is remarkably balanced, with each city generating over \$300,000. Brooklyn holds the top position with \$319,705.48, followed closely by Queens and Manhattan. This minimal variance suggests consistent market demand and effective sales performance across all geographical locations.

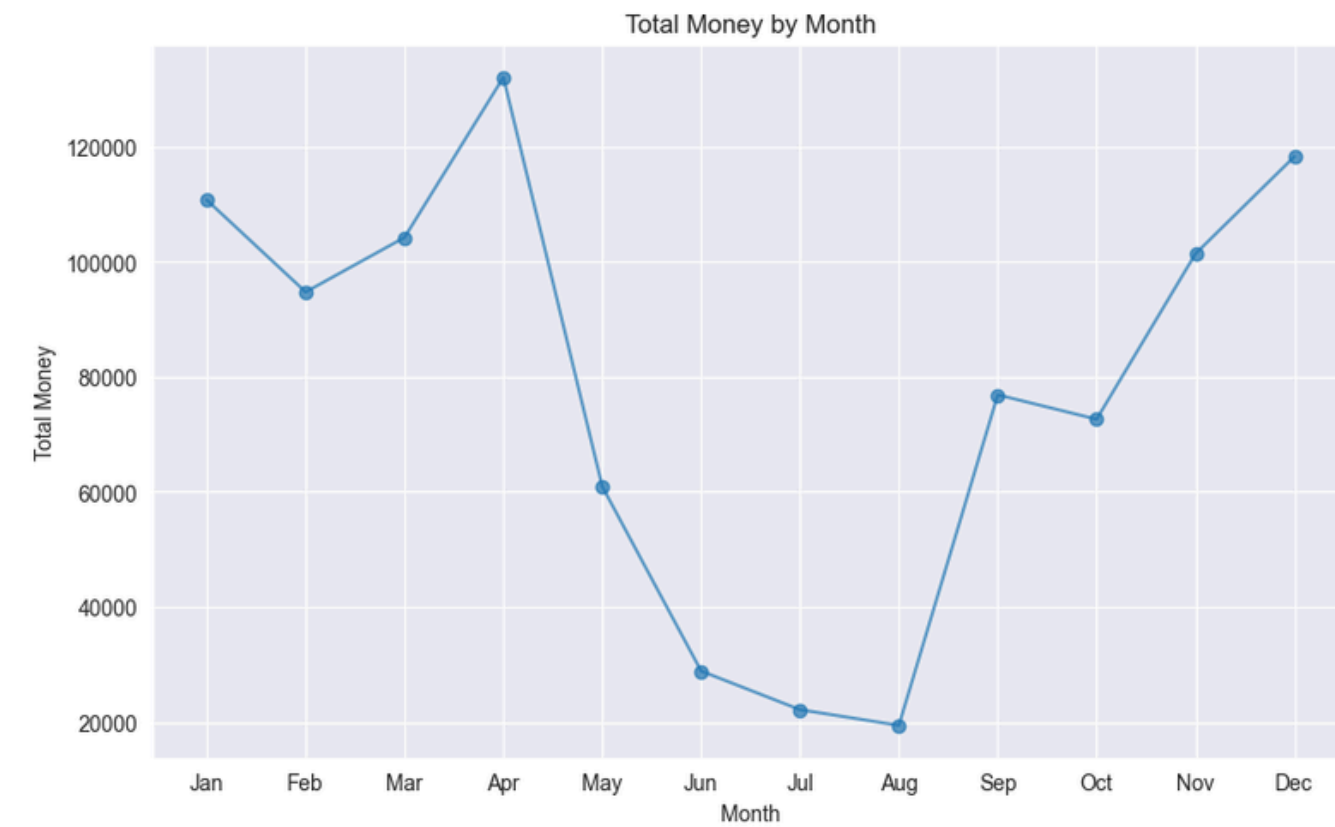
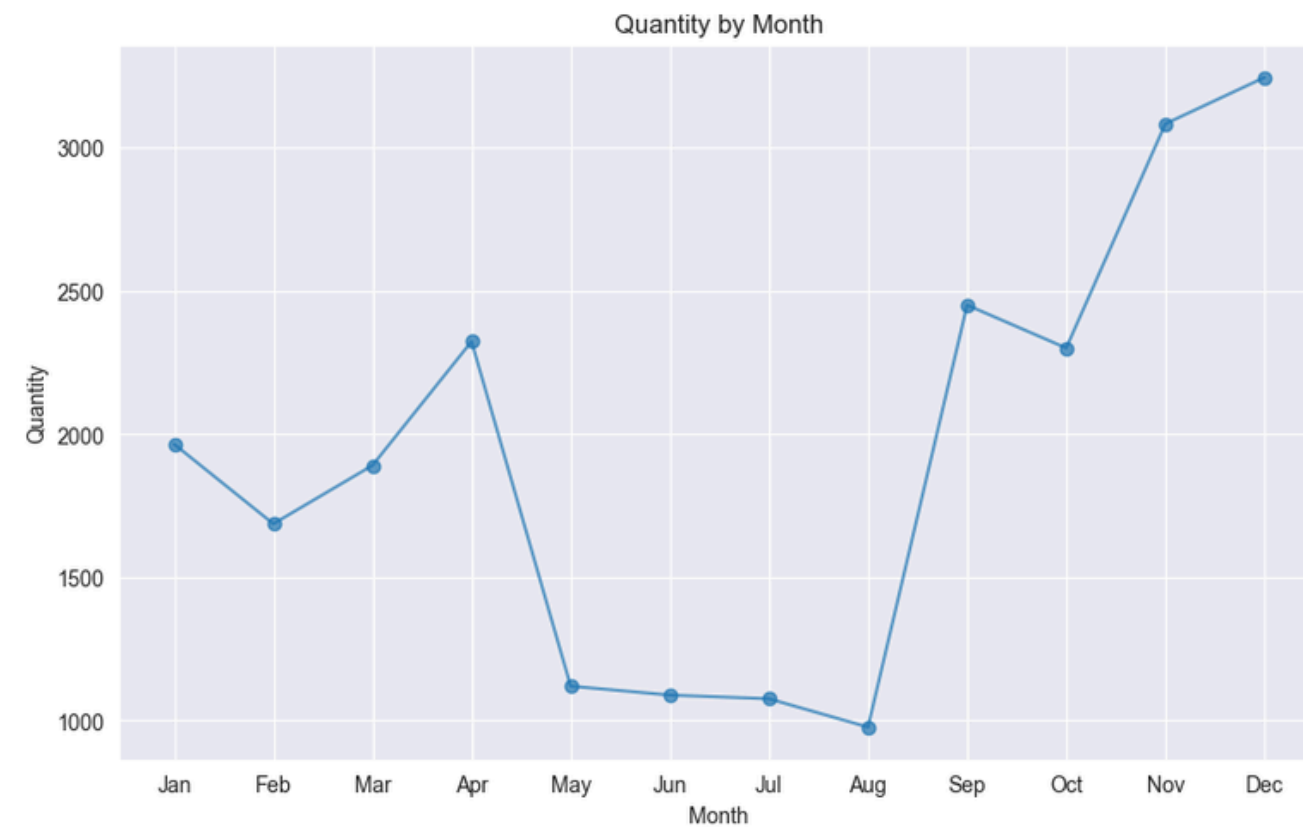
	Branch	TotalMoney
0	Brooklyn	319705.48
1	Queens	312842.82
2	Manhattan	309996.55



8. THE SALES TREND OVER THE MONTHS?

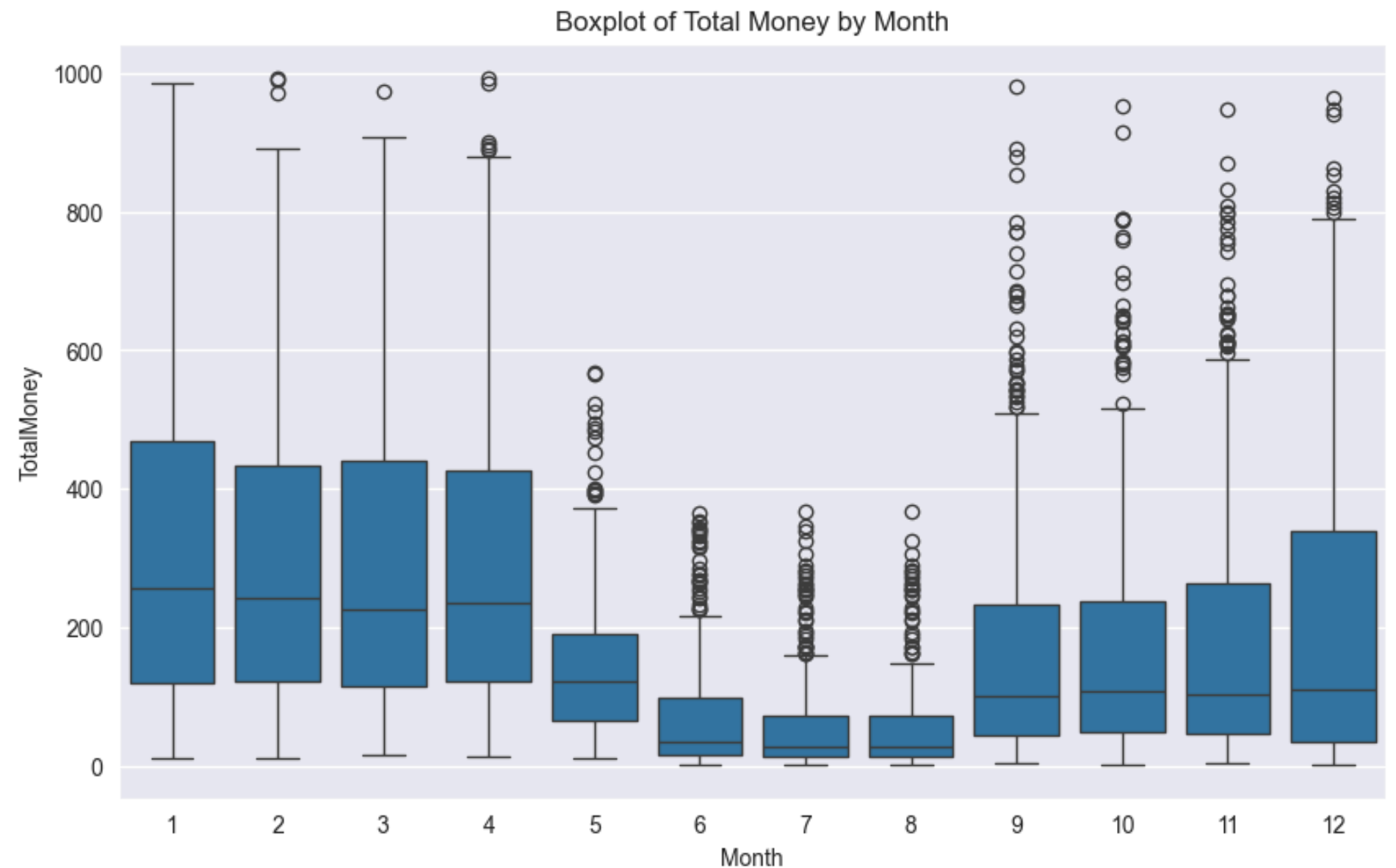
- The highest volume of transactions occurs at the end of the year, in December and November
- The lowest transaction seen in February
- The chart shows a positive trend toward the end of the year.



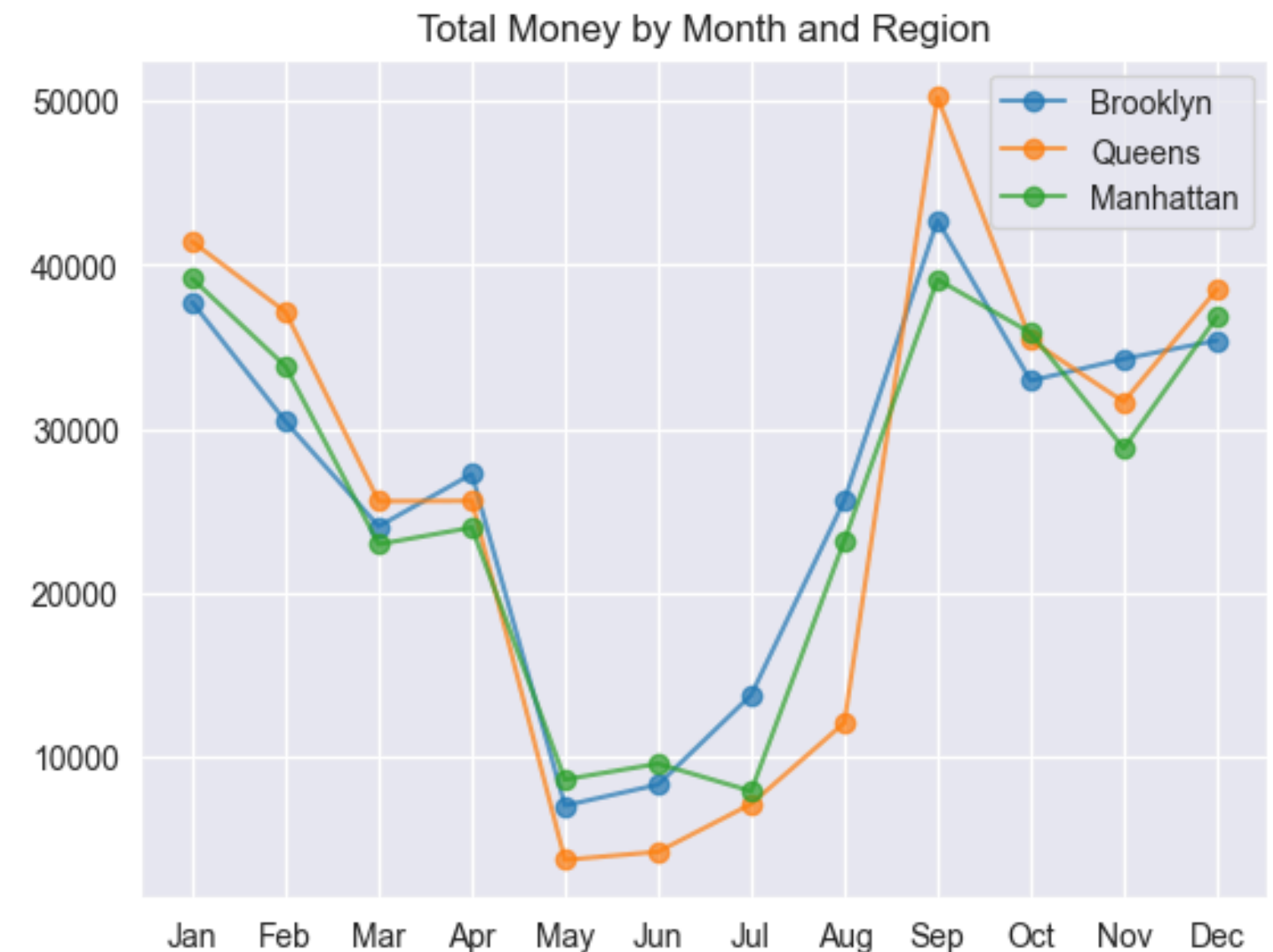


- In April, revenue was very high, even though the number of items sold was not as high as in December. This indicates that April likely sold more expensive products.
- In November and December, the number of items sold increased drastically, but the revenue did not increase as much as the increase in items.

- In January-April, The median appears to be the highest, in the range of 200-250. This means that at the beginning of the year, the average value of each transaction tends to be higher.
- in May-Augustus, most transactions were of low activity
- In September-December, the number of transactions increased drastically. Here, we can see many outliers that boosted total revenue.

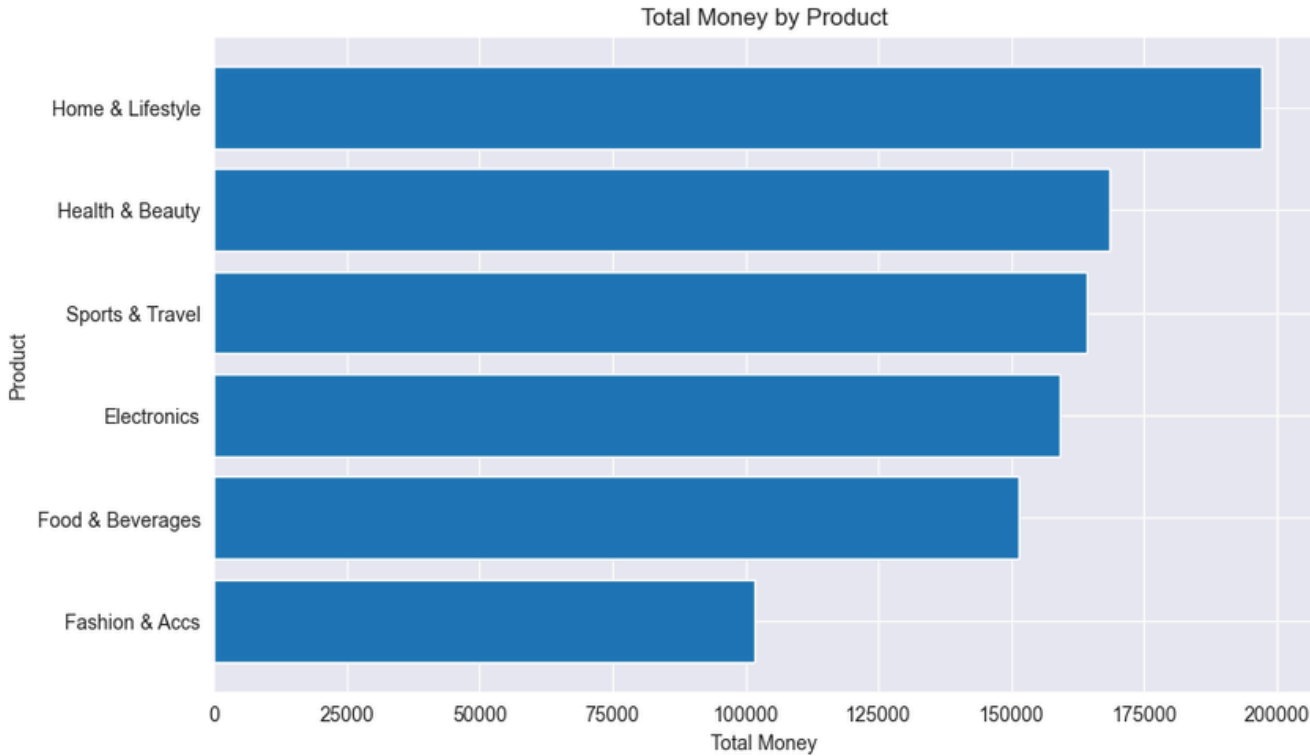


- While all regions follow a similar seasonal pattern, Queens shows the most fluktuativ, hitting the highest overall peak in September (over 50,000) despite having the lowest performance during the May-July period.
- Brooklyn demonstrated more resilience during the summer slump, maintaining higher revenue in July and August compared to the other two region.



9.TOP-EARNING PRODUCTS

	Product line	TotalMoney
0	Home & Lifestyle	197260.56
1	Health & Beauty	168570.61
2	Sports & Travel	164252.94
3	Electronics	159259.10
4	Food & Beverages	151357.73
5	Fashion & Accs	101843.91



- Customers spend the most money on Home and lifestyle
- Fashion may need a stronger promotional strategy or pricing evaluation.
- The Health & Beauty, Sports & Travel, Electronics, and Food & Beverages categories have figures that are not too far apart from each other.

10. BUSINESS STRATEGY RECOMMENDATIONS

- Maximize sales of more expensive products in April and the beginning of the year (January–April).
- Ensuring optimal stock availability for the Food & Beverages and Health & Beauty categories.
- Improving customer retention programs in the Brooklyn area
- Conducting an aggressive marketing campaign in Queens during September to capitalize on its peak sales, as well as looking for ways to stabilize performance during the low period in May–July

**THANK YOU FOR
YOUR TIME**



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DATA ANALYST PRESENTATION
